

Lezione 4 – Studi di funzione

Matematica del continuo

Modulo 11 - Studio di funzioni

Unità didattica 2 – Funzioni con logaritmi ed esponenziali

Enrico Spoletini

Università degli Studi di Milano - SSRI - CdL Online

$$y = (x+1)e^{x/(x-1)} \quad (-\infty, 1) \cup (1, +\infty)$$

$y(0) = 1$
 $y = 0 \quad x = -1$

$\lim_{x \rightarrow -\infty} y = -\infty \quad -\infty e^{\frac{-\infty}{-\infty}}$
 $\uparrow$ 1° ordine

$\lim_{x \rightarrow +\infty} y = +\infty \quad +\infty \cdot e^{\frac{+\infty}{+\infty}}$

As. Obliquo ??

$\lim_{x \rightarrow 1^-} y = 0^+ \quad (1+1)e^{\frac{1}{0^-}} \quad 2e^{-\infty}$
 $\lim_{x \rightarrow 1^+} y = +\infty \quad (1+1)e^{\frac{1}{0^+}} \quad 2e^{+\infty}$

$x = 1 \quad e^- \quad \text{As. vert. alle } dx$

$$y = (x+1) e^{\frac{x}{x-1}}$$

$$\lim_{x \rightarrow \pm\infty} \frac{y}{x} = e$$

$$\lim_{x \rightarrow \pm\infty} (y - ex) = 2e$$

$$(x+1) e^{\frac{x}{x-1}} - ex = \underbrace{x e^{\frac{x}{x-1}} - xe}_{\downarrow} + \underbrace{e^{\frac{x}{x-1}}}_{\uparrow e}$$

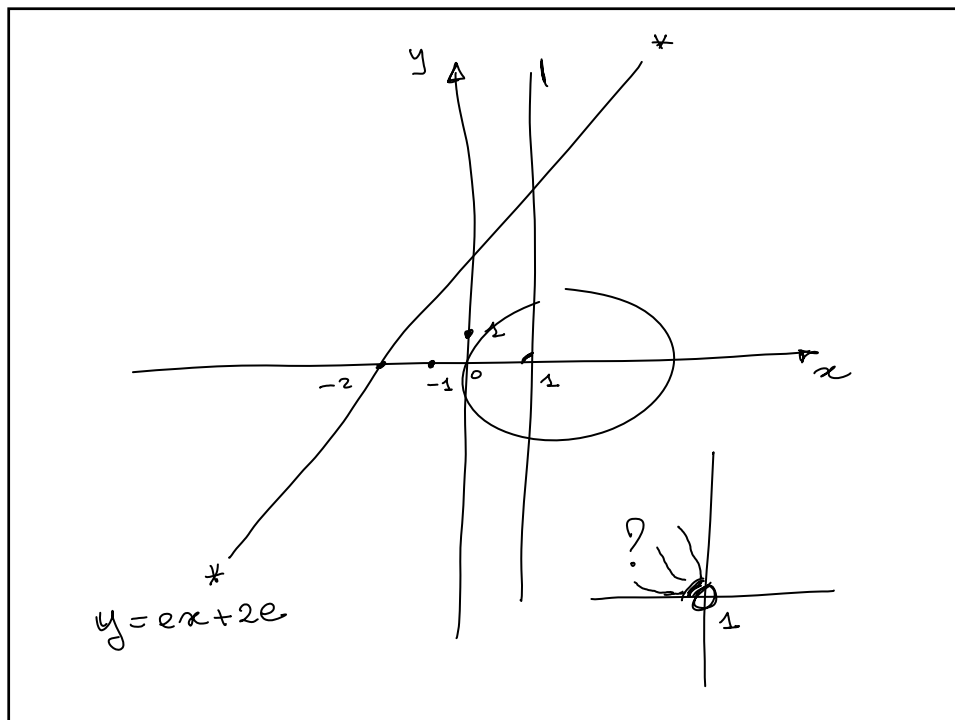
$$xe \left(\underbrace{\frac{x}{x-1} - 1}_{\sim} \right)$$

$$xe \frac{x - x + 1}{x-1} \downarrow e$$

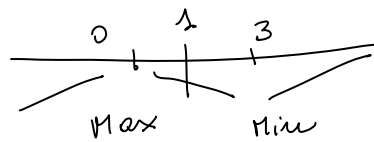
for $x \rightarrow \pm\infty$

$$y = ex + 2e$$

AS. OBL.

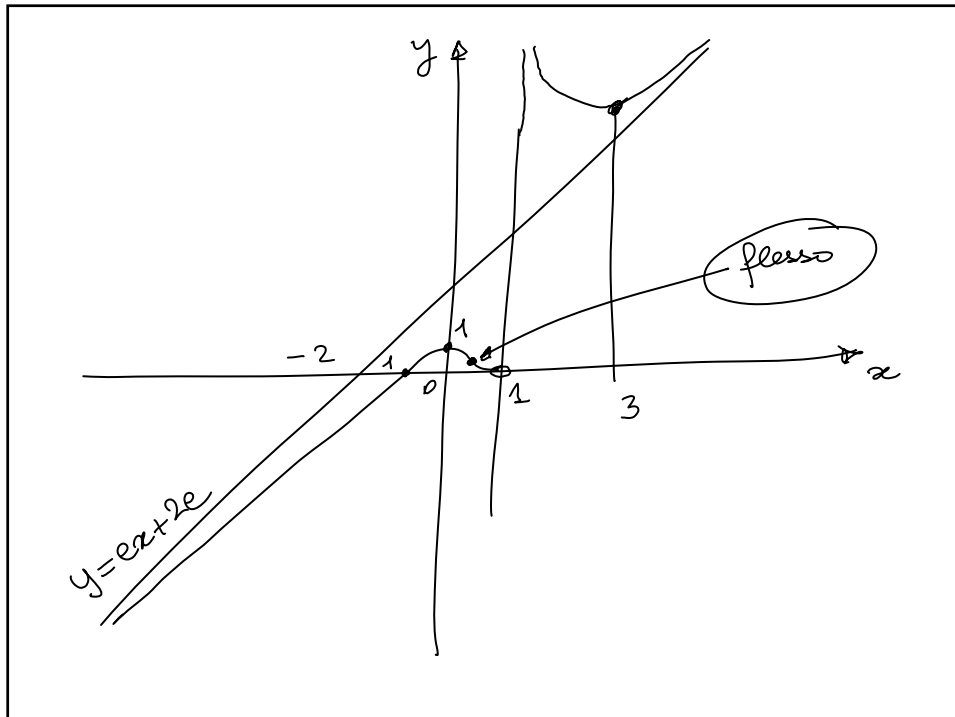


$$\begin{aligned}
 y &= (x+1)e^{\frac{x}{x-1}} \\
 y' &= e^{\frac{x}{x-1}} + (x+1) \cdot e^{\frac{x}{x-1}} \cdot \frac{(x-1) - x}{(x-1)^2} \\
 &= e^{\frac{x}{x-1}} \left(1 - \frac{x+1}{(x-1)^2} \right) = \\
 &= \frac{e^{\frac{x}{x-1}}}{(x-1)^2} \left((x-1)^2 - (x+1) \right) = \\
 &= \frac{e^{\frac{x}{x-1}}}{(x-1)^2} x(x-3) \geq 0 \quad x(x-3) \geq 0
 \end{aligned}$$



$$y(3) = 4e^{3/2}$$

$\frac{e^{-\infty}}{0^+}$	$\frac{0^+ \text{ exp.}}{0^+ \text{ pot.}}$	0^+
---------------------------	---	-------



$$\begin{aligned}
 y' &= e^{\frac{x}{x-1}} \left(1 - \frac{x+1}{(x-1)^2} \right) \\
 y'' &= \left(e^{\frac{x}{x-1}} \right)' \cdot \frac{1}{(x-1)^2} \left(1 - \frac{x+1}{(x-1)^2} \right) + \\
 &\quad \left(e^{\frac{x}{x-1}} \right) \left(- \frac{(x-1)^2 - (x+1) \cdot 2(x-1)}{(x-1)^4} \right) = \\
 &= \frac{-e^{\frac{x}{x-1}}}{(x-1)^4} \left((x-1)^2 - (x+1) + (x-1)^2 - 2(x^2-1) \right) \\
 &\quad \cancel{x^2 - 2x + 1} - \cancel{x - 1} + \cancel{x^2 - 2x + 1} - 2\cancel{x^2} + 2 = \\
 &= -5x + 3 \qquad y'' \geq 0 \qquad 5x - 3 \geq 0 \\
 &\quad \downarrow \qquad \uparrow \\
 &\quad \quad \quad \frac{\quad}{3/5} \qquad x = 3/5 \quad \text{FINE} \\
 &\quad \quad \quad \text{flush}
 \end{aligned}$$